Exchange-Only Quantum Discrete Diffusion for Language Modeling

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Abstract

We build a discrete diffusion language model whose only quantum operation is the Heisenberg exchange interaction, the native gate of semiconductor spin qubits. Each token is a single spin excitation among V qubits. The uniform D3PM corruption process is realised exactly as a random spin-shuffling (SWAP) channel, and the denoiser is an exchange circuit whose pulse areas are set by the noisy context. Measuring the circuit samples the clean token by the Born rule, so ancestral sampling needs a single shot per token and step. We show that exchange acts on the token register through the dynamical Lie algebra $\mathfrak{u}(V-1)$: the W state is dark, and every output obeys an amplitude sum rule. Nevertheless, every categorical distribution remains reachable, and nearest-neighbour hardware needs depth of at least $V-1$. Using an exact simulator for 32-qubit registers, we train character-level models on 2×10^6 characters and compare them with classical denoisers that use identical features. The exchange readout reaches 3.38 ± 0.01 (linear controller) and 3.18 ± 0.01 (MLP controller) bits/char, about 0.25 bits/char behind softmax read-outs (3.13 and 2.92). In return, the one-hot register makes relaxation and single readout flips detectable. Under exactly simulated noise, a 1% readout error costs 0.012 bits/char with a Hamming-weight check versus 0.29 for a binary code. With the same check, relaxation affecting 80% of shots costs only 0.03 bits/char. The model is classically simulable by construction, so we claim no quantum advantage; all code and results are released.

1 Introduction

Diffusion models [Sohl-Dickstein et al., 2015, Ho et al., 2020] have been extended to discrete data by replacing Gaussian noise with Markov transition matrices on a finite vocabulary [Austin et al., 2021, Hoogeboom et al., 2021], and discrete diffusion is now a competitive non-autoregressive approach to language modeling [Lou et al., 2024, Sahoo et al., 2024]. In parallel, several quantum versions of diffusion models have been proposed, in which either the data are quantum states or the denoiser is a parameterised quantum circuit [Zhang et al., 2024, Parigi et al., 2023, Cacioppo et al., 2023, Kölle et al., 2024]. These proposals generally assume a universal gate set with arbitrary single-qubit rotations.

This paper asks a hardware-driven question: *how far can a discrete diffusion language model go if the only operation available is the Heisenberg exchange interaction?* Exchange is the native two-qubit interaction of semiconductor spin qubits [Loss and DiVincenzo, 1998, Burkard et al., 2023]. "Exchange-only" architectures avoid fast single-spin control altogether and recover universality by encoding logical qubits in several spins [DiVincenzo et al., 2000, Weinstein et al., 2023]. Here we take a different route. We do not try to recover universality. Instead, we pick a data encoding in which exchange alone is already the natural set of operations for a categorical distribution over tokens.

Idea. A token from a vocabulary of size V is stored as a single spin excitation among V qubits (one-hot or unary encoding). Exchange conserves the number of excitations, so every exchange circuit acts inside the V-dimensional single-excitation sector. There it implements a unitary "mixing" of tokens. A random exchange (SWAP) channel corrupts tokens exactly as the uniform D3PM forward process does. A trained, context-dependent exchange circuit followed by a computational-basis measurement samples the denoised token by the Born rule.

Contributions. (i) We formulate an exchange-only quantum discrete diffusion model (QXDM) with a one-hot token register, an exchange-channel forward process, a Born-rule denoiser and measurement-based ancestral sampling that needs *one shot per token per step*. (ii) We characterise its expressivity. The dynamical Lie algebra on the token register is $\mathfrak{u}(V-1)$, and the W state is a dark state. This imposes an exact amplitude sum rule. Despite this, every categorical distribution remains reachable as a single Born-rule output, and nearest-neighbour hardware has a light-cone depth requirement. (iii) We give an exact and efficient simulator (single-excitation dynamics are classically simulable) and use it to train character-level models on 2×10^6 characters. We compare them with classical denoisers that share the same features. (iv) We study realistic exchange-only noise. Charge noise, dephasing, relaxation and readout errors are all simulated exactly. The one-hot encoding turns relaxation and single readout flips into *detectable* errors through a Hamming-weight check.

We stress at the outset that the model class studied here is classically simulable by construction, so we make

no claim of quantum advantage. Our goal is a hardware-native, verifiable baseline. It shows what an exchange-only device can and cannot express when it is used as the stochastic core of a generative language model.

2 Background

Uniform discrete diffusion. Let $x_0 \in \{1, \dots, V\}^L$ be a sequence. The uniform forward process corrupts each token independently, $q(x_t \,|\, x_{t-1}) = \mathrm{Cat}\big(\alpha_t e_{x_{t-1}} + (1-\alpha_t)\mathbf{1}/V\big)$, so that $q(x_t|x_0) = \mathrm{Cat}(\bar\alpha_t e_{x_0} + (1-\bar\alpha_t)\mathbf{1}/V)$ with $\bar\alpha_t = \prod_{s\le t}\alpha_s$ [Austin et al., 2021, Hoogeboom et al., 2021]. We use $T = 32$ steps, a cosine schedule and $\bar\alpha_T = 0$, so x_T is exactly uniform. A denoiser $p_\theta(x_0|x_t)$ defines the reverse kernel

$$p_\theta(x_{t-1}|x_t) = \sum_{\hat x_0} p_\theta(\hat x_0|x_t)\, q(x_{t-1}|x_t, \hat x_0), \qquad (1)$$

factorised over positions. The negative evidence lower bound (NELBO) is $\sum_{t\ge 2}\mathbb{E}\,\mathrm{KL}\big(q(x_{t-1}|x_t,x_0)\,\|\,p_\theta(x_{t-1}|x_t)\big) - \mathbb{E}\log p_\theta(x_0|x_1)$. The prior term vanishes because $\bar\alpha_T = 0$. Eq. (1) is exactly the law of "draw $\hat x_0 \sim p_\theta(\cdot|x_t)$, then draw x_{t-1} from the analytic posterior". This is what makes a sampling-only (Born-rule) denoiser natural.

Exchange interaction. For spins a, b the isotropic exchange Hamiltonian is $H_{ab} = \frac{J}{4}\boldsymbol\sigma_a\cdot\boldsymbol\sigma_b$, which equals $\frac{J}{2}\mathrm{SWAP}_{ab}$ up to a constant. A pulse therefore implements

$$E_{ab}(\theta) = e^{-i\theta\,\mathrm{SWAP}_{ab}} = \cos\theta\,\mathbb{1} - i\sin\theta\,\mathrm{SWAP}_{ab}, \qquad (2)$$

with $\theta = \int J\,dt/2$. Exchange commutes with every global spin rotation, so it conserves both the total spin S^2 and the magnetisation S_z [DiVincenzo et al., 2000, Marvian, 2022].

3 Exchange-only quantum discrete diffusion

3.1 Token register

Each sequence position owns a register of V qubits. The token k is stored as $|e_k\rangle = |0\cdots010\cdots0\rangle$ (excitation on qubit k). Let $\mathcal{H}_1 = \mathrm{span}\{|e_k\rangle\}$.

Lemma 1 (Exchange on the token register)**.** *$E_{ab}(\theta)$ leaves $\mathcal{H}_1$ invariant. On $\mathcal{H}_1$, $E_{ab}(\theta) = e^{-i\theta}\exp(i\theta L_{ab})$, with the edge Laplacian $L_{ab} = (e_a - e_b)(e_a - e_b)^\top$. Explicitly, up to the global phase the gate acts as $\left(\begin{smallmatrix} u & v \\ v & u\end{smallmatrix}\right)$ on (e_a, e_b), where $u = e^{i\theta}\cos\theta$ and $v = -ie^{i\theta}\sin\theta$, and as the identity on all other e_k.*

Proof. SWAP_{ab} permutes $e_a \leftrightarrow e_b$ and fixes the other e_k, so on $\mathcal{H}_1$ we have $\mathrm{SWAP}_{ab} = \mathbb{1} - L_{ab}$. Since $L_{ab}^2 = 2L_{ab}$, $e^{i\theta L_{ab}} = \mathbb{1} + \frac{e^{2i\theta}-1}{2}L_{ab}$, which gives the stated entries. □

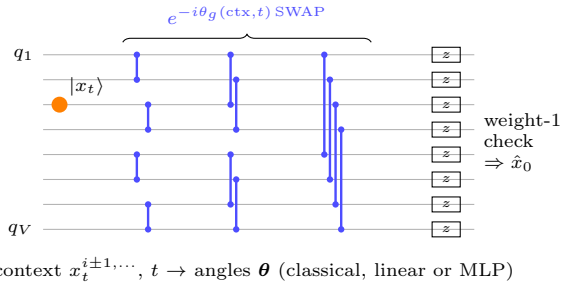


Figure 1: Exchange-only denoiser for one token. The noisy token x_t^i is loaded as a single excitation on V qubits. Layers of exchange pulses with context-dependent angles (here a butterfly network, i.e. $\log_2 V$ layers per block) act inside the single-excitation sector. A z-basis measurement with a Hamming-weight check returns $\hat x_0 \sim |\langle \hat x_0|U|x_t^i\rangle|^2$, which is fed to the analytic posterior $q(x_{t-1}|x_t, \hat x_0)$.

Thus a circuit of G exchange pulses is simulated exactly in $O(G)$ time per basis input. Every gate updates two amplitudes: $\psi_a \leftarrow \psi_a - \Delta$ and $\psi_b \leftarrow \psi_b + \Delta$, with $\Delta = (\sin^2\theta - i\sin\theta\cos\theta)(\psi_a - \psi_b)$. We checked Lemma 1 against a full 2^n-dimensional state-vector simulation with the Heisenberg Hamiltonian (Table 1).

3.2 Forward process as an exchange channel

Proposition 1. *Let $\Phi_t(\rho) = (1-\beta_t)\rho + \beta_t\,\mathbb{E}_{\pi\sim S_V}[P_\pi \rho P_\pi^\dagger]$, where P_π permutes qubits according to a uniformly random permutation. With $\beta_t = 1 - \alpha_t$, the induced transition on one-hot states is the uniform D3PM kernel $q(x_t|x_{t-1})$.*

Proof. Every permutation is a product of transpositions, i.e. full-SWAP exchange pulses $E_{ab}(\pi/2)$ up to phase. $P_\pi\,|e_k\rangle = |e_{\pi(k)}\rangle$, and $\pi(k)$ is uniform over $\{1..V\}$ for a uniform π. □

The forward process is therefore physically an exchange ("spin-shuffling") channel. Training never has to run it on hardware, because $q(x_t|x_0)$ has a closed form.

3.3 Exchange-circuit denoiser

For position i we load $|e_{x_t^i}\rangle$, apply $U_{\boldsymbol\theta} = \prod_g E_{a_g b_g}(\theta_g)$ and measure. Hence

$$p_{\boldsymbol\theta}(\hat x_0 = k \,|\, x_t, t) = \big|\,\langle e_k|U_{\boldsymbol\theta(x_t,t)}|e_{x_t^i}\rangle\,\big|^2. \qquad (3)$$

The pulse angles are set by a classical controller from sparse features of the *noisy* context: the tokens at offsets $-4..4$, five neighbour pairs, the time step t, and the joint feature (t, x_t^i). In the *linear* controller, $\theta_g = J_g(t)\sum_{f\in\mathrm{active}} W_{f,g}$, i.e. a sum of learned per-feature pulse areas. The *exchange envelope* $J_g(t)$ scales all pulses at noise level t. When $J \to 0$ we get $U \to \mathbb{1}$, which implements "copy x_t" at low noise. The nonlinear (Born-rule)

map from angles to probabilities is performed entirely by the exchange circuit. As an ablation we also use *data re-uploading* [Pérez-Salinas et al., 2020]: each active feature drives its own exchange block and the blocks are applied in sequence. To isolate the read-out, we further pair the circuit with a two-layer *MLP controller*. This is the same network as our classical MLP baseline, but its output is G angles instead of V logits.

We use two connectivities. A *butterfly* network pairs $k \leftrightarrow k \oplus 2^s$ in layer $s = 0, \ldots, \log_2 V - 1$; the default is 3 blocks, i.e. 15 layers and 240 gates for $V = 32$. A *1D chain* uses nearest-neighbour brick-wall layers, which suits linear quantum-dot arrays.

3.4 Sampling and training

Measurement-based ancestral sampling. At step t every position runs its circuit once and is measured. The classical posterior $q(x_{t-1}|x_t^i, \hat{x}_0^i)$ is then sampled, and the next round re-prepares $|e_{x_{t-1}^i}\rangle$. Because Eq. (1) is a mixture over $\hat{x}_0$, a single shot per token per step samples the model exactly. No amplitude or probability estimation is required. Generating L tokens costs LT circuit executions of depth D on V qubits, and positions are independent within a step.

Training. We minimise the D3PM hybrid objective $T\,\mathcal{L}_{t-1} + \lambda\,\mathrm{CE}(x_0)$ with a single random t per sequence and $\lambda = 0.01$ [Austin et al., 2021]. In simulation we differentiate exactly. On hardware, the generator SWAP has eigenvalues ± 1, so the two-term parameter-shift rule $\partial_\theta f = f(\theta + \frac{\pi}{4}) - f(\theta - \frac{\pi}{4})$ applies to every exchange angle [Mitarai et al., 2018, Schuld et al., 2019]. We verified it numerically (Table 1).

4 Expressivity of exchange on the token register

Proposition 2 (Dark state and sum rule)**.** *For any exchange circuit, $U|W\rangle = |W\rangle$ (up to the removed global phase), where $|W\rangle = V^{-1/2}\sum_k |e_k\rangle$. Consequently $\sum_k \langle e_k|U|e_x\rangle = 1$ for every input x.*

Proof. $L_{ab}\mathbf{1} = 0$ for every edge, so each factor $e^{i\theta L_{ab}}$ fixes $|W\rangle$. Since U is unitary, $\langle W|\,U = \langle W|$, and $\sum_k \langle e_k|U|e_x\rangle = \sqrt{V}\,\langle W|U|e_x\rangle = \sqrt{V}\,\langle W|e_x\rangle = 1$. □

Physically, $|W\rangle$ lies in the maximal-spin multiplet ($S = V/2$). The remaining $V-1$ states of $\mathcal{H}_1$ have $S = V/2 - 1$, and SU(2)-invariant exchange cannot couple the two sectors.

Proposition 3 (Dynamical Lie algebra)**.** *For a connected interaction graph, the Lie algebra generated by $\{iL_{ab}\}$ on $\mathcal{H}_1$ is $\mathfrak{u}(V-1)$, acting on $|W\rangle^\perp$ and trivially on $|W\rangle$.*

We verified Proposition 3 numerically by computing the Lie closure for chains and butterflies with $V \in \{4, 6, 8\}$ (Table 1). In every case the dimension equals $(V-1)^2$, not V^2.

Proposition 4 (Every distribution is reachable)**.** *For any input x and any distribution p on $\{1..V\}$, there is a U in the group generated by exchange with $|\langle e_k|U|e_x\rangle|^2 = p_k$ for all k.*

Proof. By Prop. 3, U acts as an arbitrary unitary on $|W\rangle^\perp$. Hence the reachable outputs are exactly the unit vectors ψ with $\langle W|\psi\rangle = \langle W|e_x\rangle = V^{-1/2}$, i.e. $\|\psi\| = 1$ and $\sum_k \psi_k = 1$. Write $\psi_k = \sqrt{p_k}e^{i\varphi_k}$. Phases with $\sum_k \sqrt{p_k}e^{i\varphi_k} = 1$ exist if and only if segments of lengths $\{\sqrt{p_k}\} \cup \{1\}$ close a polygon. This holds because $\sum_k \sqrt{p_k} \geq \sum_k p_k = 1$ and $\sqrt{p_j} \leq 1 \leq 1 + \sum_{k\neq j}\sqrt{p_k}$. □

The sum rule therefore does not restrict the set of output distributions. It does restrict *how* they are parameterised: the amplitudes are coupled through a shared phase constraint, unlike independent softmax logits. Reachability also needs enough depth. On a nearest-neighbour chain, an excitation travels at most one site per layer, so with $D < V - 1$ layers some outputs have zero amplitude. Table 2 confirms this. Random Dirichlet targets are fitted to a median KL below 10^{-5} nats by a single butterfly block (5 layers) and by a 32-layer chain, but a 16-layer chain leaves more than 10 nats. Conversely, the low dimension of the dynamical Lie algebra ($(V-1)^2$, polynomial in V) rules out the exponential concentration associated with barren plateaus [McClean et al., 2018, Ragone et al., 2024].

5 Noise and error detection

Noise in exchange-only spin qubits mainly comes from four sources [Burkard et al., 2023]: **(a)** quasi-static charge noise on J, i.e. $\theta \to \theta(1+\epsilon)$; **(b)** dephasing from fluctuating local fields, a random Z phase per qubit and layer; **(c)** energy relaxation $|1\rangle \to |0\rangle$ with qubit-dependent rate γ_k per layer; and **(d)** readout bit flips with probability q. Channels (a) and (b) preserve $\mathcal{H}_1$. Channel (c) maps $\mathcal{H}_1$ to the vacuum, and (d) acts on classical outcomes. All of them can therefore be simulated *exactly* for $V = 32$ qubits: we use Monte-Carlo averaging over (a, b), the no-jump trajectory for (c), and closed forms for (d).

Hamming-weight check. Any valid outcome has Hamming weight 1, so checking the weight is a free symmetry verification [Bonet-Monroig et al., 2018, McArdle et al., 2019]. A relaxation event yields weight 0 and is always detected. Post-selection then returns the normalised no-jump state $\propto \mathrm{diag}(\sqrt{1-\gamma_k})^D\,U\,|e_x\rangle$. In particular, *uniform* relaxation ($\gamma_k \equiv \gamma$) costs only a reduced acceptance rate $(1-\gamma)^D$ and leaves the model distribution unchanged. For readout flips, a single flip gives weight 0

or 2 and is detected. An undetected error needs two flips, which happens with probability $(V-1)q^2(1-q)^{V-2} = O(Vq^2)$. Compare a dense binary code on $\log_2 V$ qubits: there every outcome is valid, and errors occur undetected with probability $1-(1-q)^{\log_2 V} \approx q \log_2 V$.

6 Experiments

6.1 Setup

Data. We use the English `fortunes` collection, i.e. the data files of the Debian `fortunes`/`fortunes-min` packages, excluding `ascii-art` and attribution lines. It is lower-cased and mapped to $V = 32$ symbols (`a--z`, space, `. , ' ? !`). Entries are shuffled and split 90/5/5, giving 2.01×10^6 training, 1.19×10^5 validation and 1.09×10^5 test characters. The unigram entropy is 4.26 bits.

Models. All local denoisers share the featuriser of Sec. 3.3 (F = 6,831 sparse features, 17 active per position). They differ only in the read-out. *Log-linear softmax*: summed feature logits with a per-t logit scale (the classical analogue of $J(t)$). *MLP softmax*: summed 240-dimensional feature embeddings, two GELU layers, softmax. *Exchange (linear ctrl.)*: summed pulse areas for $G = 240$ exchange gates (butterfly, 3 blocks) with envelope $J(t)$. *Exchange (MLP ctrl.)*: the MLP above with a 240-angle output feeding the same circuit.

Training and evaluation. Each run uses Adam [Kingma and Ba, 2015], learning rate 2×10^{-3} (200-step warm-up, cosine decay), 10^4 steps, and 64 windows of 64 characters per step. Training is on CPU: about 25 minutes per exchange model on one core, using our real-arithmetic simulator. We report the test NELBO in bits/char, summing $\mathcal{L}_{t-1}$ over *all* $T = 32$ steps (no Monte-Carlo over t) on 256 test windows. For sample quality we draw 256 unconditional samples of length 64 by measurement-based ancestral sampling. We score them by (a) their bits/char under an interpolated Kneser–Ney 5-gram fitted to the training split [Chen and Goodman, 1999] and (b) the fraction of interior words of length ≥ 2 that occur in the training vocabulary.

6.2 Verification of the simulator and theory

Table 1 confirms Lemma 1 against a full state-vector simulation with the Heisenberg Hamiltonian. Leakage is exactly zero and total spin is conserved. The dynamical Lie algebra has dimension $(V-1)^2$ in every case (Prop. 3). The dark-state and sum-rule identities hold to machine precision, and the $\pi/4$ parameter-shift rule reproduces automatic differentiation. Table 2 supports Prop. 4 and shows the light-cone failure of shallow nearest-neighbour circuits.

Table 1: Numerical checks. Top: exact agreement of the single-excitation simulator with a full 2^6-dimensional Heisenberg simulation of a 4-layer chain, together with the identities used in Secs. 3–4 (for $V = 32$). Bottom: dimension of the dynamical Lie algebra generated by $\{iL_{ab}\}$.

max. amplitude error vs. full 2^6 state vector	3.5×10^{-16}
max. leakage out of $\mathcal{H}_1$	0
$\|[U, S^2]\|_{\max}$	1.1×10^{-15}
$\|U\|W\rangle - \|W\rangle\|_{\max}$ (Prop. 2)	0
sum rule $\|\sum_k \langle e_k\|U\|e_x\rangle - 1\|$	5.0×10^{-16}
parameter shift ($\pm\pi/4$) vs. autograd	7.1×10^{-17}

graph	V	$\dim \mathfrak{g}$	$(V-1)^2$	V^2
chain	4	9	9	16
chain	6	25	25	36
chain	8	49	49	64
butterfly	4	9	9	16
butterfly	8	49	49	64

Table 2: Column reachability (Prop. 4): KL divergence (nats) after fitting a single circuit column to 5 random Dirichlet(α) targets on $V = 32$ outcomes. The 16-layer chain is below the light-cone depth $V - 1$.

circuit	α	median KL	max KL
butterfly 1	0.1	1.5e-10	1.9e-07
butterfly 3	0.1	3.6e-10	2.5e-06
chain 16	0.1	1.1e+01	4.7e+01
chain 32	0.1	1.6e-08	1.1e-05
butterfly 1	1.0	4.0e-09	6.1e-08
butterfly 3	1.0	9.2e-06	3.1e-04
chain 16	1.0	2.0e+01	4.0e+01
chain 32	1.0	6.4e-16	1.6e-07

6.3 Language modeling

Table 3 gives the main result. Seed-to-seed variation is below 0.01 bits/char. The exchange denoiser with a linear controller reaches 3.375 bits/char. A log-linear softmax on the same features reaches 3.129, with 7× fewer parameters. Giving both the same nonlinear controller narrows the absolute numbers but not the gap: 3.177 for the exchange read-out vs. 2.919 for softmax. Sample-quality metrics give the same ranking. The MLP softmax produces the most word-like text, while both exchange models and the log-linear model produce text with about a third valid words and 5-gram bits/char between 4.6 and 5.0. For comparison, i.i.d. unigram text scores 6.41 and real text 2.22. No model collapses: the unigram entropy of every sample set lies between 4.19 and 4.24 bits (data: 4.27).

6.4 Ablations

Table 4 isolates the design choices. (i) *Depth.* One butterfly block (5 layers) gives 3.59. Gains diminish beyond 3 blocks: 2 blocks give 3.41, 3 give 3.38 and 6 give 3.36. (ii)

Table 3: Test NELBO (bits/char; mean ± s.d. over seeds) and sample quality (seed 0, temperature 1). *Gen. bits* is the bits/char of generated text under a Kneser–Ney 5-gram (lower is more plausible), and *words* is the percentage of generated words found in the training vocabulary. For reference, the Kneser–Ney 5-gram itself reaches 2.11 bits/char on the test split and the unigram entropy is 4.26 bits.

denoiser (same features)	params	seeds	test NELBO ↓	gen. bits ↓	words (%) ↑
Exchange (linear ctrl.)	1.65M	3	3.375 ± 0.006	5.02	32.5
Exchange (MLP ctrl.)	1.76M	3	3.177 ± 0.005	4.65	36.1
Log-linear softmax	0.22M	3	3.129 ± 0.001	4.57	36.2
MLP softmax	1.70M	3	2.919 ± 0.003	3.80	52.9
Held-out data	–	–	–	2.22	95.4
i.i.d. unigram	–	–	–	6.41	11.5

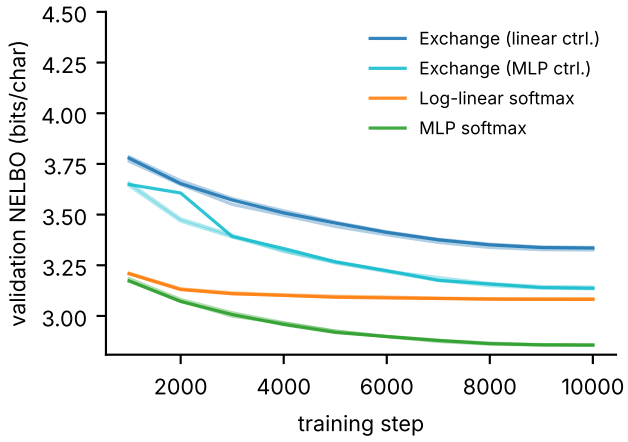


Figure 2: Validation NELBO during training (all seeds).

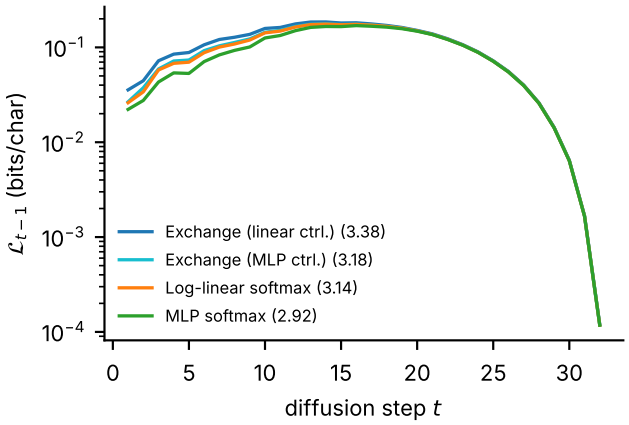


Figure 3: Per-step contribution $\mathcal{L}_{t-1}$ to the test NELBO (seed 0; totals in the legend). The exchange read-out loses at every noise level, with the largest relative gap at small t, where the denoiser must copy x_t with high confidence.

t exchange denoiser (linear ctrl.), one sample

```
32 flmi!jlfpaqbmhn,gzsu?fsul,k,'osexxjrx?awnjjh!'wzba,riyntpkzi?hdg
28 ga.zejfwjdd usm?laoscwagaxfstzeghpogn.glrbtnzjrd?fzblzt,?mk.gn.,
24 m,kzfgf jmd  vmocao cd,gzattozo!h'o! .nvabyxzwr?!flgl,yu?ak.'nk,
20 mmkzf f ?mda vr.oao ag,ufztoozm!v'a! .n!a sxanrc!floleyjimb.'skw
16 mmkwfcf lvda i..'ao agyufttsgzj!v aa hn!a s anrn!flo,eetimb.'skw
12 mpke cf lcda i. 'a, a oufttssek!e da hn!a ' an ,hfly,petimb.'s w
 8 mpte mf lida is 'a, a ountessekie da kn a ' an thlly,petimb.'s w
 4 mpte sf lida is 'a, a qunte sekre dacre a b an thely petimb.'s w
 0 mpte st lipa is 'a, a aunte sekre dacre a bjan thely metimb.'s w
```

Figure 4: Measurement-based reverse diffusion from uniform noise ($t = 32$) to text ($t = 0$), shown every 4 steps.

Connectivity. A 16-layer nearest-neighbour chain is below the light-cone depth and degrades to 3.91. A 48-layer chain (3× the gates) matches the butterfly (3.35). (iii) *Envelope.* Removing $J(t)$ costs 0.024 bits/char. (iv) *Objective.* Training with the x_0 cross-entropy alone instead of the hybrid NELBO objective costs 0.20 bits/char. (v) *Features.* Without pair features the exchange model loses 0.27 and the log-linear model 0.28, so the two read-outs use context features to a similar extent. (vi) *Re-uploading* with one 80-gate block per feature (16 blocks in sequence) reached 3.58 in half the steps. This matches the additive circuit with the same block size (3.59), but at 16× the circuit depth.

6.5 Hardware noise

Figure 5 shows the four noise channels of Sec. 5. *Charge noise*: a 2% relative angle error costs 0.004 bits/char and 5% costs 0.03. *Dephasing* up to 0.03 rad per layer is negligible (+0.003). *Relaxation*: with the weight check the NELBO changes by less than 10^{-3} bits/char up to $\bar{\gamma} = 10^{-2}$ per layer, and rises by only 0.03 at $\bar{\gamma} = 0.1$, where 80% of shots are rejected. For uniform rates it is exactly unchanged (3.3672 at every γ), as predicted. Without the check the same relaxation costs 0.07 at $\bar{\gamma} = 10^{-3}$ and 7.3 bits/char at 0.1. *Readout*: at $q = 10^{-2}$ the one-hot register with weight check loses 0.012 (acceptance 73%), while a binary code loses 0.29. At $q = 3 \times 10^{-2}$ the losses are 0.15 and 0.91.

7 Discussion

What the exchange read-out costs. With identical features and controllers, replacing the softmax by an exchange circuit plus a Born-rule measurement costs about 0.25 bits/char (linear controller: 3.38 vs. 3.13; MLP con-

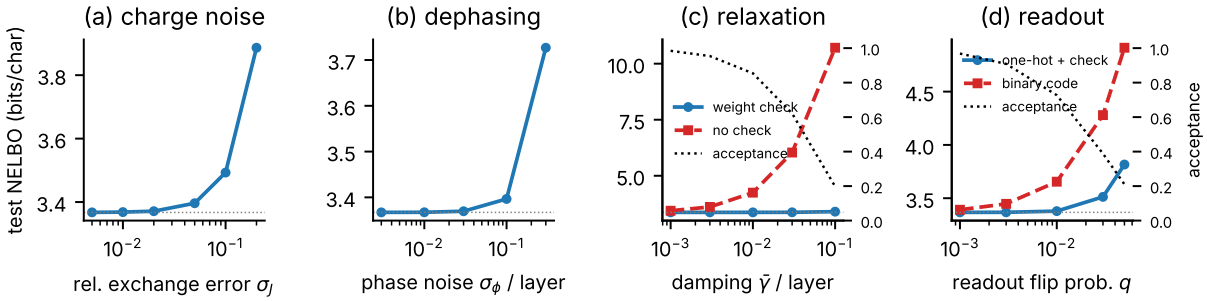


Figure 5: Test NELBO of the trained exchange denoiser (seed 0, 64 test windows, clean value 3.367 dotted) under exactly simulated noise on all 32 qubits of every register. (a) Quasi-static relative error of every exchange angle. (b) Gaussian Z-phase per qubit per layer. (c) Relaxation with qubit-dependent rates $\gamma_k \in [0.5, 1.5]\bar{\gamma}$ per layer: Hamming-weight check with post-selection (blue; dotted: acceptance) vs. no check, where a lost excitation is read as a random token (red). (d) Readout bit flips: one-hot register with weight check (blue; dotted: acceptance) vs. the same distribution read out through a dense $\log_2 V$-qubit binary code (red).

Table 4: Ablations (seed 0, test NELBO in bits/char). G = exchange gates, D = layers.

variant	G	D	NELBO
Default: butterfly, 3 blocks (15 layers, 240 gates)	240	15	3.375
Log-linear, no pair features	–	–	3.411
1D chain, 16 brick-wall layers	248	16	3.913
1D chain, 48 brick-wall layers	744	48	3.350
Butterfly, 1 block(s)	80	5	3.586
Butterfly, 2 block(s)	160	10	3.405
Butterfly, 6 block(s)	480	30	3.358
Data re-uploading (1 block/feature, 5000 steps)	80	5	3.578
No exchange-pulse envelope $J(t)$	240	15	3.399
No pair features	240	15	3.642
Trained with x_0 cross-entropy only	240	15	3.571

troller: 3.18 vs. 2.92). This happens even though the circuit has more parameters, and Proposition 4 guarantees that every output distribution is reachable. The gap is therefore one of *parameterisation and optimisation*, not of reachability. A softmax combines feature contributions additively in log space, i.e. as a product of experts. Additive pulse areas instead interfere through a deep, periodic, non-commuting map, and the sum rule of Prop. 2 couples all output amplitudes. The per-step decomposition (Fig. 3) shows where the loss is incurred. Depth helps but saturates (Table 4). Data re-uploading, which composes feature blocks non-commutatively, matched but did not beat additive angles with the same block size, at much larger depth. The exchange envelope $J(t)$, which lets the device turn its pulses off at low noise, gives a small but consistent gain (0.024 bits/char).

What the exchange read-out buys. The read-out is native. Sampling requires one shot per token per step, and the one-hot register makes the dominant hardware errors *detectable*. Relaxation, the most damaging channel without a check, becomes almost harmless with post-selection. It is exactly harmless when rates are uniform. A single readout flip can never silently change a token. For realistic readout errors ($q \approx 10^{-2}$) the one-hot register loses 0.01 bits/char, compared with 0.29 for a dense binary code (Fig. 5). The model is also insensitive to percent-level exchange miscalibration and to dephasing below 0.1 rad per layer. These are regimes reported for current silicon devices [Burkard et al., 2023].

Hardware cost. One token per denoising step needs $V = 32$ spins and, in the default butterfly, 15 exchange layers (240 pulses). On a linear dot array the light cone forces at least $V - 1$ brick-wall layers, and the 48-layer chain (744 pulses) matches the 240-pulse butterfly in quality. Positions are conditionally independent within a step, so L registers run in parallel and generation takes T rounds of measure-and-reset. Butterfly connectivity needs long-range exchange (e.g. via shuttling).

Limitations. (i) *Classical simulability.* A single excitation evolves like a single particle, so every result here can be reproduced classically in $O(VG)$ time; this is how we trained. We make no claim of quantum advantage. A natural extension that leaves the classically simulable regime is to let $k > 1$ tokens share one register. The Hamming-weight-k sector has dimension $\binom{V}{k}$ and is still preserved by exchange. (ii) *Local context.* All denoisers here see a fixed window of ± 4 noisy characters plus pair features. Absolute numbers are therefore far from autoregressive models with longer context: an interpolated Kneser–Ney 5-gram reaches 2.11 bits/char on the same split [Chen and Goodman, 1999]. We also trained a bidirectional Transformer denoiser with global context. Within our CPU

budget it stalled at the "copy x_t + unigram" solution (test NELBO 4.23 bits/char). We report it only in the released logs, because it is not a well-tuned reference. (iii) *Small scale.* One corpus (`fortunes`, 2×10^6 characters), $V = 32$ and $T = 32$ steps. Uniform-transition diffusion is known to trail absorbing-state (masked) diffusion for text [Austin et al., 2021, Sahoo et al., 2024]. An exchange-only analogue of the absorbing state, for example a dedicated "mask" qubit, is a natural next step. (iv) *Noise models.* Noise was injected at inference time into a model trained noise-free. We did not model crosstalk, leakage out of the qubit manifold, or state-preparation errors.

8 Conclusion

A discrete diffusion language model can be built from the exchange interaction alone. Tokens are single spin excitations, corruption is spin shuffling, and denoising is a context-controlled exchange circuit read out by the Born rule. The exchange group on the token register is $\mathrm{U}(1) \oplus \mathrm{U}(V-1)$, fixed by spin symmetry. It is small enough to avoid barren plateaus, yet large enough to reach every categorical distribution. At equal features the exchange read-out trails a softmax by about 0.25 bits/char. In return it comes with native one-shot sampling and built-in detection of relaxation and readout errors. We hope these exact, reproducible baselines help calibrate claims about quantum generative models of language.

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